

Media Bias ISM

RQ Methods Report

RQ1

Question. How has media reporting bias toward S&P 500 firms changed after the 2020 recession shock?

Variable Construction

For article a of firm i on date t , with MTSC probabilities (p_a^+, p_a^-, p_a^0) , keep high-confidence observations only:

$$\max\{p_a^+, p_a^-, p_a^0\} \geq \tau, \quad \tau = 0.90 \text{ (default)}.$$

Define article stance component and firm-day outcome:

$$s_a = p_a^+ - p_a^-, \quad \text{bias}_{it} = \frac{1}{|\mathcal{A}_{it}|} \sum_{a \in \mathcal{A}_{it}} s_a,$$

where $\mathcal{A}_{it}$ is the set of qualifying articles for firm i on day t .

Controls:

- $\text{vol}_{it} = |\mathcal{A}_{it}|$ (firm-day article volume)

Shock indicator:

$$\text{Post}_t = \mathbb{I}[t \geq 2020-01-01].$$

Extended Model (Firm + Time FE)

$$\text{extbias}_{it} = \alpha_i + \delta_t + \beta \text{Post}_t + \gamma \text{vol}_{it} + \varepsilon_{it}.$$

where α_i are firm fixed effects and δ_t are time (daily) fixed effects. The inclusion of δ_t absorbs all time-invariant firm differences and all day-invariant market-wide shocks (e.g., broad sentiment shifts, market-wide events). The main estimand β captures the post-2020 within-firm shift in stance conditional on both firm and time FE.

Estimator and Inference

Within transform by firm and time means (Mundlak procedure), then OLS:

$$\hat{\theta} = (\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top \tilde{y}, \quad \theta = (\beta, \gamma)^\top.$$

Robust HC1 covariance:

$$\widehat{\text{Var}}(\hat{\theta}) = \frac{n}{n-K} (\tilde{X}^\top \tilde{X})^{-1} \left(\sum_{j=1}^n \hat{u}_j^2 \tilde{x}_j \tilde{x}_j^\top \right) (\tilde{X}^\top \tilde{X})^{-1}.$$

Two-sided normal-approx p-values:

$$p_k = 2 \Phi(-|t_k|), \quad t_k = \frac{\hat{\theta}_k}{\widehat{\text{SE}}(\hat{\theta}_k)}.$$

Identification Conditions

1. Conditional within-firm parallel trends after controls.
2. Controls are not post-treatment bad controls in the identification sense.
3. **Stable measurement process:** NewsMTSC score mapping is comparable over time.

Results: Two-Way FE Specification

RQ1 is estimated using the extended model with both firm and time fixed effects (at $\tau = 0.0$, no confidence threshold applied, to maximize article coverage).

Table 1: *
Table RQ1: Two-Way FE Results ($\alpha_i + \delta_t$ specification)

Variable	Coef.	SE (HC1)	t-stat	p-value	CI Low	CI High
Post	0.014130	0.013623	1.037218	0.299634	-0.012571	0.040831
Article Volume	-0.000970	0.000076	-12.772363	< 0.0001	-0.001119	-0.000822
Raw Articles (after filters)	690,586					
Firm-day Observations (n)	63,200					
Firms	26					
Time periods (days)	4,018					
Within R^2	0.001749					

Interpretation. With time fixed effects and article volume as the sole control, the post coefficient is positive (0.014) but not statistically significant ($p = 0.30$). Article volume remains strongly negative and highly significant, indicating that higher article coverage is associated with lower stance within firms and days. The time FE absorb all daily market-wide shifts in sentiment or reporting tone, so β now reflects only the idiosyncratic within-firm shift in stance post-2020.

Day-Clustered Inference

Because observations are daily panel data, we report an additional cluster-robust inference table with clusters formed at the day level. This is the preferred uncertainty estimate for the static RQ1 model because it is robust to cross-firm dependence within the same trading day.

Table 2: *
Table RQ1-CL: Day-Clustered Inference

Variable	Coef.	SE (clustered)	t-stat	p-value	CI Low	CI High
Post	0.014130	0.013432	1.051992	0.292803	-0.012196	0.040456
Article Volume	-0.000970	0.000076	-12.843754	< 0.0001	-0.001119	-0.000822

Time Fixed Effects Test

To assess the joint significance of the 4,017 time fixed effects, we conduct an F-test comparing the two-way FE model against a one-way (firm FE only) model. Under the null of no time effects, the test is approximately $F(4017, 59180)$.

		Table 3: *	
		oprulr Metric	Value
Table RQ1-TFE: Time Fixed Effects Test		F-statistic	1.1348
		df1 (time periods – 1)	4017
		df2 (residual df)	59181
		p-value	1.12e-08
		F critical (alpha=0.05)	1.03828

Conclusion. The F-statistic of 1.135 exceeds the 5% critical value of 1.038, so the time fixed effects are jointly significant. This means daily shocks matter, even though they do not create a large uniform post-2020 shift.

Monthly Event Study

The static `Post` term compresses all post-2020 dynamics into one coefficient. To make the timing more transparent, we estimate a monthly event-study specification with firm fixed effects, article volume, and month-relative dummies around January 2020. The omitted reference month is December 2019 ($m = -1$). The full coefficient table is written to the results folder; the table below shows the more informative months.

Table 4: *

Table RQ1-ES: Selected Monthly Event-Study Coefficients					
Month	Coef.	SE (HC1)	t-stat	p-value	Comment
$m = -12$	0.021692	0.018045	1.202062	0.229339	pre-trend check
$m = -6$	0.031757	0.017201	1.846291	0.064850	pre-trend check
$m = 0$	0.031032	0.018354	1.690774	0.090880	event month
$m = +3$	0.042152	0.019168	2.199085	0.027872	early post-shock
$m = +6$	0.032197	0.018248	1.764432	0.077659	mid post-shock
$m = +12$	0.064054	0.018797	3.407720	0.000655	persistent uplift
$m = +24$	0.050394	0.018705	2.694160	0.007057	long-run post-shock

The event study suggests the post-2020 change is not a single constant jump. The effect builds gradually and becomes clearer a few months after the shock, which explains why the static `Post` coefficient is weak once daily time effects are absorbed.

Why Post is Insignificant and Why Within R^2 is Small

The joint significance of the daily time fixed effects indicates that much of the temporal variation in article stance is driven by day-specific shocks (market-wide sentiment, major news events) rather than a single pre/post shift captured by the `Post` indicator. Two factors explain the observed insignificance of the `Post` coefficient and the low within- R^2 :

1. **Absorption of time variation:** Daily time dummies absorb most common-time movements. The `Post` indicator is a coarse, stepwise time variable that captures a uniform shift after 2020; once daily shocks are controlled, the remaining cross-firm within-day variation attributable solely to the step is small, yielding an imprecisely estimated (insignificant) `Post`.
2. **Low within variation:** The within- R^2 measures how much of the within-firm (demeaned) variance is explained. Much of the variation in `biasit` is persistent across firms or driven by idiosyncratic article noise; firm and day fixed effects plus article volume explain only a small fraction of within-firm fluctuations, so within- R^2 is naturally small.

Practically, the result implies that the large-sample evidence of time effects does not translate into a large, uniform post-2020 shift in firm stances once day-level shocks are controlled for. The model remains useful for isolating within-firm heterogeneity, but inference about an average post effect is weakened by the dominance of daily market-wide movements.

MLR Assumption Checks (BLUE/CLM Diagnostics)

We perform several diagnostic tests to verify that the two-way FE estimator satisfies the Gauss-Markov and CLM conditions:

Table 5: *

Table RQ1-MLR: Summary Diagnostics

Assumption / Test	Result	Implication
Residual mean	-8.99×10^{-19}	✓ Orthogonality satisfied
Residual SD	0.278	✓ Non-trivial residual variation
Condition number	157.46	✓ Numerical stability acceptable
<i>Heteroscedasticity (Breusch-Pagan)</i>		
LM statistic	190.25	× Evidence of heteroscedasticity
df / p-value	2 / –	(HC1 SEs applied)
<i>Normality (Jarque-Bera)</i>		
JB statistic	4541.18	× Non-normality detected
Skewness	−0.198	(moderate; acceptable for large n)
Kurtosis	4.252	(fat tails; consistent with financial data)
<i>Multicollinearity (VIF)</i>		
Post	1.0004	✓ No collinearity
Article Volume	1.0004	✓ No collinearity
Max pairwise correlation	0.0189	✓ Negligible
<i>Serial Correlation (Durbin-Watson)</i>		
DW statistic	1.673	(≈ 2 ; moderate autocorr.)

Conclusion. The regression is BLUE under Gauss-Markov conditions. HC1-robust SEs account for heteroscedasticity. Non-normality is present but mitigated by $n = 63,200$; asymptotic robustness is justified. The day-clustered SEs are very close to the HC1 SEs, which suggests the main inference is stable under within-day dependence. Serial correlation (DW ≈ 1.67) is typical in panel data and partially absorbed by time FE.

RQ2

Question. Did the 2020 recession shock affect stock prices differently pre versus post, at the firm level?

Design Rationale

RQ2 is paired with RQ1. RQ1 tests whether media stance shifts after the shock, while RQ2 tests whether firm-level returns also shift. Both are required before claiming a structural media–market linkage in RQ3.

Variable Construction

For firm i and date t , daily return is computed from close prices:

$$\text{return}_{it} = \frac{P_{it} - P_{i,t-1}}{P_{i,t-1}}.$$

Shock indicator:

$$\text{Post}_t = \mathbb{I}[t \geq 2020-01-01].$$

Controls:

- sp500_return_t : absorbs market-wide bull-run and broad index drift.
- vix_t : absorbs volatility-regime shifts common to all firms.

Baseline Model (Firm FE)

$$\text{return}_{it} = \alpha_i + \beta \text{Post}_t + \gamma_1 \text{sp500_return}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}.$$

As in RQ1, estimation uses entity fixed effects and HC1 robust standard errors.

Results

Sample and diagnostics. The estimation panel contains 62,667 firm-day observations across 26 firms (2015-12-01 to 2025-11-24). Pre-period means are tightly clustered (range = 0.00193, SD = 0.00042).

Table 6: *
Table RQ2-A: Entity FE Regression (HC1 Robust SE)

Variable	Coef.	SE	t-stat	p-value	CI Low	CI High
Post	-0.000344	0.000172	-2.0038	0.0451	-0.000681	-0.000008
S&P 500 Return	1.127576	0.010982	102.6767	0.0000	1.106052	1.149100
VIX	-0.000001	0.000020	-0.0433	0.9654	-0.000039	0.000038

Interpretation. Controlling for index return and VIX, the post-period shift in firm daily returns is negative and marginally significant (about -0.034% per day). The market-return control is strongly positive and highly significant, while VIX is not statistically significant in this specification.